MIE237 – Project Report

University of Toronto

Beats vs Books: Music's Impact on Arithmetic Accuracy and Speed

Team 9	
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1.0 Introduction

A heavily debated question, particularly among students, is whether listening to music while working or studying helps improve focus and performance. Many students use background music while completing schoolwork, yet its effects on cognitive performance are not always clear. Prior literature suggests that the effects of background music vary due to several factors, notably task demand and cognitive load [1]. Because of this, the relationship between background music and task performance remains an important topic for investigation.

This study examines whether background music influences arithmetic performance, and whether that influence changes depending on task difficulty. More specifically, the study compares performance on easy and hard mental arithmetic tasks under two audio conditions: music and no music. Performance was evaluated using two measures: the number of problems attempted and the accuracy of responses.

The overall goal of this experiment is to better understand how an everyday study habit like listening to music may influence performance on mentally demanding tasks. Since arithmetic problems require attention, working memory, and fast processing, they are a useful way to test whether background music is beneficial, harmful, or dependent on the level of cognitive load. Finally, this research aims to contribute to a better understanding of how study environments can affect academic task performance among undergraduate students.

2.0 Literature Review

The paper by Goltz & Sadakata notes that the evidence on background music (BGM) and cognitive task performance is mixed, and that results vary based on the music type, task, and individual characteristics of participants [1]. The paper examines how people use BGM “optimally” during everyday cognitive tasks such as reading, writing, critical thinking, and

memorization, and it measures key factors such as preferred music types and frequency, as well as participant characteristics including their age [1]. Two main findings from this paper are specifically relevant to the design of our study. Firstly, participants reported that they tend to use less BGM when focused on more difficult tasks, indicating that cognitive demand can affect whether BGM is considered disruptive or beneficial [1]. This supports our decision to compare easy versus hard arithmetic when testing how BGM affects accuracy and response time. Secondly, a negative correlation between age and BGM frequency was found in the study, suggesting that younger generations typically use BGM more frequently when working or studying compared to older adults [1]. This provides further justification for using music rather than background speech as the audio distraction type for our study design with our group of undergraduate participants, as music is a common 'study habit' in this age group.

In Study 1 of *The Effects of Background Music on Primary School Pupils' Task Performance,* Hallam, Price, and Katsarou examined the effects of background music on arithmetic task performance in children aged 10-12 [2]. The students were given sixth grade coursework involving fraction manipulation, and performance was evaluated using two measures: the number of arithmetic problems attempted within a fixed 15-minute time interval and accuracy. The students performed these tasks under conditions with and without background music. The results demonstrated that students completed more problems when background music was present (mean = 34.9 problems, SD = 7.7) compared to without music (mean = 27.3 problems, SD = 7.8; p= 0.02). However, no statistically significant difference in accuracy was observed, with mean accuracy rates of 84% with background music present and 80% without.

These results suggest that background music increased problem-solving speed, but did not affect accuracy. Although the participant population in the study by Hallam, Price, and Katsarou

consisted of primary school students, the study's task structure of time-constrained mental arithmetic relies on sustained attention and active working-memory engagement, making the study conditions transferable to an undergraduate engineering context.

3.0 Methods

This section describes the participants, variables of study, experimental design, and hypotheses.

3.1 Participants

Participants in this study were undergraduate students at the University of Toronto. The experiment used a within-subject design, meaning that there were twelve participants and each completed the arithmetic task under both audio conditions: background music present, and no background music. Using a within-subject design allowed performance to be compared across conditions within the same participant, helping reduce the influence of individual differences in arithmetic ability.

3.2 Study Variables

This study uses two independent variables and two dependent variables.

3.2.1 Independent Variables

1) **Background music presence**: there were two levels including no music (silence) and background music present at a constant volume. Background music was of the smooth jazz genre and was the same for all participants.
2) **Task difficulty**: difficulty had two levels that were defined by working-memory load.
    - **Easy** (low working-memory load): single-step arithmetic problems requiring little retention and manipulation (e.g., one-digit + two-digit addition, two-digit -

one-digit subtraction, one-digit x one-digit division, and one-digit x one-digit multiplication).

- **Hard** (high working-memory load): multi-step arithmetic that requires carrying or borrowing of results (e.g., (38 + 47) - 16, 132 - 78 + 49, 64 / 7).

3.2.2 Dependent Variables

1) **Problems Attempted:** this variable was measured as the total number of arithmetic problems within both difficulty levels completed during both fixed 1.5-minute time frames, each containing either music or silence performance conditions.
2) **Accuracy**: the percentage of correctly answered questions per condition (described below in Section 3.3).

3.3 Experimental Design

This study used a 2x2 within-subject design. Each participant completed the task under both levels of the audio condition and encountered both levels of task difficulty. As a result, performance was examined across four condition combinations: (1) music + hard, (2) music + easy, (3) no music + hard, and (4) no music + easy.

3.3.1 Apparatus

The experiment was conducted using an interactive Figma-based arithmetic task interface created by the team. This digital interface acted as the platform through which participants completed the full study. At the beginning of the experiment, participants entered their name into the interface. The system then automatically assigned each participant to start with either the music condition or the no-music condition. Before their session began, they were played a sample of the audio used during the experiment and told to adjust the volume to what they

usually study to. This was done so participants could best replicate what a music vs silent study environment looked like for them.

At the end of the task, participant response data was automatically saved as a CSV file. Participants were instructed to send this file to one team member, who compiled all participant files into one spreadsheet containing the full dataset.

3.3.2 Procedure

Each participant completed two timed 1.5-minute sessions of arithmetic questions. One round was completed with background music present, and the other was completed without music. Each round lasted 1.5 minutes. During both rounds, participants answered a mixed set of easy- and hard-level arithmetic questions (defined in Section 3.2.1). Importantly, the interface did not identify whether a question was easy or hard, which was done to reduce potential bias in how participants approached problems.

At the start of the experiment, the interface randomly assigned the participant to begin with either the music condition or the no-music condition. After the first 1.5-minute round ended, the second round automatically began under the alternate audio condition. This way, each participant completed both audio conditions, while the starting order varied across participants to reduce possible order effects such as fatigue or practice. Participant responses were recorded automatically by the system. After data collection was complete, the files from all participants were combined into a single dataset. From this dataset, the number of problems attempted and the accuracy for each participant were determined separately for each condition combination.

3.4 Hypotheses

Null and alternative hypotheses were formulated for each dependent variable prior to data collection.

3.4.1 Null Hypotheses

- H0_attempted (null hypothesis for problems attempted): there is no effect of audio condition or task difficulty on number of problems attempted.
- H0_accuracy (null hypothesis for accuracy): there is no effect of audio condition or task difficulty on accuracy.

3.4.2 Alternative Hypotheses

- H1_attempted (alternative hypothesis for problems attempted): number of attempted problems differs as a function of audio presence and/or task difficulty.
- H1_accuracy (alternative hypothesis for accuracy): accuracy differs as a function of audio type and/or task difficulty.

4.0 Analyses

The main analysis conducted in R was a blocked two-factor ANOVA test as the study used a 2x2 within-subject design. Audio condition and task difficulty were the two experimental factors, while participants were included as the blocking factor. Each of the dependent variables, number of problems attempted and accuracy, were analyzed separately.

This approach was chosen based on the two factor blocked design framework taught in lecture. Since each participant completed all four conditions and individuals may differ in baseline arithmetic abilities, treating participants as blocks allowed for comparisons to be made within the same individual, reducing between-subject variability. The model used for this analysis is:

$$Y_{ijk} = \mu + \pi_k + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

Here, Y_{ijk} represents the response variable we are measuring effects for (number of problems attempted or accuracy), π_k represents the effect of participant k, α_i represents the effect of audio

condition, β_j represents the effect of task difficulty, and $(\alpha\beta)_{ij}$ represents the interaction between audio condition and task difficulty. ϵ_{ijk} represents residual variability unexplained by the model.

First, descriptive statistics were computed for each of the four experimental conditions (no music/hard, music/easy etc.). For each dependent variable, the mean, standard deviation, and sample size was computed for each combination of audio condition and task difficulty (see sections 5.1.1 and 5.1.2).

Interaction plots were also generated to briefly assess whether the effect of task difficulty differed across audio conditions for both dependent variables. These plots were only used to provide an initial sense of whether an interaction effect might be present before formal statistical testing.

Next, two ANOVA models, one for each dependent variable, included audio condition and task difficulty as factors. Their interaction was included as fixed effects, while participants were treated as blocking factors. The interaction effect was examined to check whether the effect of task difficulty depended on audio condition. For insignificant interaction, the main effects were interpreted directly from the ANOVA p-values.

Finally, diagnostic plots were used to assess whether the assumptions of ANOVA were reasonably satisfied. Residuals vs fitted plots were checked to assess homoscedasticity and Q-Q plots were checked to see if the residuals were approximately normal.

5.0 Results

The following section includes both descriptive statistics and results from the blocked two-factor analysis of variance conducted on the compiled dataset across all 12 of our participants.

5.1 Descriptive Statistics

The following section computes descriptive statistics consisting of means and standard deviations to provide an initial comparison of outcomes across the 2x2 within-subject design.

5.1.1 Descriptive Statistics on Data Collected With Respect to Number of Problems Attempted

The following table summarizes the output of lines 40-47 in the BeatsvsBooks.R file, where the mean and standard deviation of the number of attempted problems within each of the four experimental conditions was computed.

Table 1. Summary Statistics for Number of Attempted Problems

Presence of Music	**Task Difficulty**	**Mean Number of Problems Attempted**	**Standard Deviation**	**Sample size, n**
no_music	Easy	7.5	3.80	12
no_music	Hard	7.08	3.65	12
music	Easy	7.42	2.84	12
music	Hard	6.75	2.73	12

Since this is a within-subject experiment, each of the 12 participants experienced all four conditions. Here, each row represents one of the four conditions and hence all rows have n = 12.

5.1.2 Descriptive Statistics Across Effects of Audio and Difficulty on Accuracy

The following table summarizes the output of lines 49-56 in the BeatsvsBooks.R file, where the mean and standard deviation of accuracy within each of the four experimental conditions was computed.

Table 2. Descriptive Statistics for Accuracy

Presence of Music	**Task Difficulty**	**Mean Accuracy**	**Standard Deviation**	**Sample size, n**
no_music	easy	97.4	6.05	12

no_music	hard	78.2	26.9	12
music	easy	93.7	8.81	12
music	hard	83.9	15.5	12

Since this is a within-subject experiment, each of the 12 participants experienced all four conditions (n = 12).

5.2 Inferential Statistics (ANOVA Summary)

The following table reports the p-values for the effects of audio, difficulty, as well as their interaction on both the number of attempted problems and accuracy. These values were generated from the blocked two-factor ANOVA conducted on lines 167-174 of the BeatsvsBooks.R file [Appendix A].

Table 3. Blocked Two-Factor ANOVA Results for Problems Attempted and Accuracy

Effect	**Attempted (p)**	**Accuracy (p)**	**Conclusion**
Audio	0.650	0.834	No statistically significant effects of audio condition on number of tasks attempted or accuracy ($p>0.05$).
Difficulty	0.242	0.005	Statistically significant effect of difficulty on accuracy **only ($p=0.005$)**.
Interaction	0.785	0.331	No significant interaction between audio condition and task difficulty on accuracy or number of tasks attempted.

There was a significant main effect of difficulty on accuracy, $F(1, 33) = 9.01$, $p = 0.005$, with lower accuracy observed in the hard condition. There was no significant effect of audio condition ($p = 0.834$) and no significant interaction between audio condition and difficulty ($p = 0.331$). Since no significant interaction between audio condition and difficulty was observed, the impact of difficulty on accuracy is consistent across audio conditions.

5.3 ANOVA Assumption Verification

This section analyzes the Q-Q and Residuals vs Fitted diagnostic plots to assess whether the data in the BeatsvsBooks_CompiledData.csv reasonably satisfies the assumptions of ANOVA.

5.3.1 Attempted Number of Problems Diagnostic Plots

As seen below in Figure 1, the residuals appear randomly scattered with no clear pattern, which suggests that homoscedasticity is satisfied. The Q-Q plot indicates that residuals are approximately normally distributed, with minor deviations in the tails due to a small number of outliers including points 21, 23, and 24.

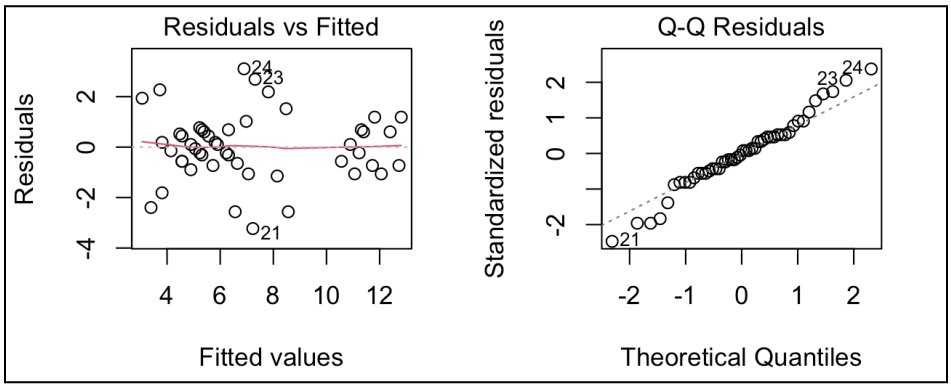


Figure 1. Attempted Number of Problems Residuals vs Fitted and QQ plots.

5.3.2 Accuracy Diagnostic Plots

The Residuals vs Fitted plot suggests a slight variation in spread, with a few outliers present (seen below in Figure 2).

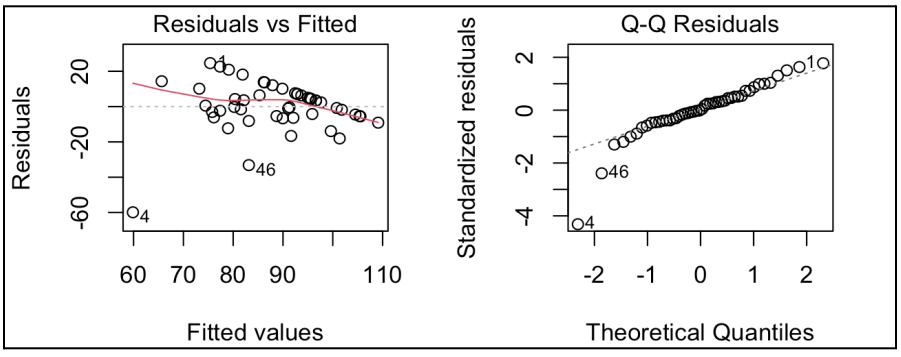


Figure 2. Accuracy Residuals vs Fitted and QQ plots

However, no strong funnel pattern is observed, and the variance appears reasonably constant overall. The Q-Q plot indicates that residuals are approximately normally distributed, with slight deviations at the tails attributable to a small number of outliers notably, points 46 and 4.

6.0 Discussion

This section discusses the implications for this experiment's results.

6.1 Conclusions

Table 4. Hypothesis-Conclusions Table

	H0_attempted	**H0_accuracy**
Conclusion	1. With respect to task difficulty → **Fail to reject** (p = 0.242) 2. With respect to audio condition → **Fail to reject** (p = 0.650)	1. With respect to task difficulty → **Reject** (p = 0.005) 2. With respect to audio condition → **Fail to reject** (p = 0.834)
Interactions: ● No significant interaction between audio condition and difficulty (p = 0.331) for accuracy ● No significant interaction between audio condition and difficulty (p = 0.785) for number of problems attempted No interaction effects were observed, indicating that the impact of difficulty is consistent across audio conditions.		

This shows that task difficulty influences accuracy, while background music does not significantly affect accuracy or the number of problems attempted. Our results partly align with the findings of Hallam, Price, and Katsarou, who also found that background music did not significantly affect accuracy [2]. However, unlike their study (which showed that music increases problem solving speed), our data showed no change in the number of problems attempted. This was likely due to our interval length (1.5-minutes), whereas theirs were longer (15 minutes).

6.2 Reasons for Results and Limitations

Using smooth jazz at a constant volume might be considered "low-arousal" for some participants who typically work under louder music. This might have caused the cognitive load to not be high

enough for observable change. In the future, selecting a music genre that is considered "higher-arousal" might have a larger impact on accuracy and attempted questions. While we made the decision to have participants adjust the music volume to what they usually study to, this may have introduced a variable that could have neutralized the music's impact across the group. Table 5 lists the possible limitations of this study that could have potentially impacted the results.

Table 5. Study Limitations

Limitation	**Justification**
Sample size	With only n = 12 participants, this study may lack the statistical power to detect smaller effects, especially regarding the number of problems attempted.
Duration	While 1.5 minutes ensured that participants would not get "burned out", cognitive interference or benefits from music might only manifest over longer periods of time such as what was done in the Hallam, Price, and Katsarou [2] study.
Interface-induced bias	Since all participants were answering the questions by typing their answer into the box, the method with which answers were recorded might have introduced interface-induced bias compared to if answers were recorded on paper or with multiple choice.

6.3 Recommendations For Further Study

As mentioned previously, increasing the sample size, lengthening the collection length/process, and using a different genre of music might help show a stronger relationship for both H0_accuracy and H0_attempted. More formally, increasing the testing window to 10-20 minutes might help increase the long term effect of listening to music while studying. Additionally, finding a genre of music, such as pop or rock, that has an increased effect on cognitive load may yield more accurate results.

7.0 References

[1] F. Goltz and M. Sadakata, “Do you listen to music while studying? A portrait of how people use music to optimize their cognitive performance,” *Acta Psychologica*, vol. 220, no. 103417, p. 103417, Oct. 2021, doi: https://doi.org/10.1016/j.actpsy.2021.103417.

[2] S. Hallam, J. Price, and G. Katsarou, “The Effects of Background Music on Primary School Pupils’ Task Performance,” *Educational Studies*, vol. 28, no. 2, pp. 111–122, Jun. 2002, doi: https://doi.org/10.1080/03055690220124551.

Appendix A: ANOVA Summaries and Inferential Statistics (Section 5.3)

R output - ANOVA summary for Number of Attempted Problems

Attempted

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Audio_Condition	1	0.5	0.52	0.210	0.650
Difficulty	1	3.5	3.52	1.417	0.242
Participant_ID	11	395.1	35.91	14.450	1.24e-09 ***
Audio_Condition:Difficulty	1	0.2	0.19	0.075	0.785
Residuals	33	82.0	2.49		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R output - ANOVA summary for Accuracy

	Df	Sum	Sq Mean	Sq F value	Pr(>F)
Audio_Condition	1	13	12.5	0.045	0.83374
Difficulty	1	2521	2520.7	9.008	0.00509 **
Participant_ID	11	2635	239.5	0.856	0.58898
Audio_Condition:Difficulty	1	271	271.3	0.970	0.33194
Residuals	33	9234	279.8		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Appendix B: Attribution Table

	ALEXIS	GABRIELA	FRANCESCA
SIGNATURES			
1.0 Introduction	WD, ET, FP		FP
2.0 Literature Review	WD, ET, FP		ET, FP
3.0 Methods	WD, ET, FP		MR, ET, FP
3.1 Participants	WD, ET, FP		FP
3.2 Study Variables	WD, ET, FP		MR, ET, FP
3.3 Experimental Design	WD, MR, FP		MR, ET, FP
3.4 Hypotheses	WD, ET, FP		FP
4.0 Analyses	ET, FP	WD, ET, FP	
5.0 Results	ET, FP	WD, ET, FP	
5.1 Descriptive Statistics	ET, FP	WD, ET, FP	
5.2 Inferential Statistics (ANOVA Summary)	ET, FP	WD, ET, FP	
5.3 ANOVA Assumption Verification	FP	WD, ET, FP	
6.0 Discussion	ET, FP	ET, FP	WD, ET, FP

6.1 Conclusions	FP	ET, FP	WD, ET, FP
6.2 Limitations	FP		WD, ET, FP
6.3 Recommendations for Further Study	FP		WD, ET, FP
References	WD		
Appendix	WD	WD	WD

WD – Wrote Draft

MR – Major Revision

ET – Edited

DC – Data Collection

GC – General Contribution (for tasks that don't involve writing and editing)

FP – Final Proofread of COMPLETE DOCUMENT for flow and consistency

Appendix C: Declaration of Original Authorship

Those who have signed their names below confirm that

- *the contents of the attached report are entirely original, written by the identified author or team of authors of the report;*
- *any ideas or expression of ideas taken from other sources have been accurately cited and referenced according to the referencing standard specified by the course*
- *they have read the portion, excerpted on page 2 of this document, of the Calendar of the Faculty of Applied Science and Engineering and that they understand its definition of plagiarism; and, moreover,*
- *they understand that if they provide another student with any part of their own or their team's work, for whatever reason, and the student having received the work uses it for the purposes of committing an academic offence, the person providing the work will be considered an equal party in the offence and will be subject to academic sanctions.*

Student Number	Print Name	Signature	Date
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1011043620	Gabriela Dos Santos		04/07/2026
1010951864	Francesca Mangano		04/07/2026

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- *(ii) does or omits to do anything for the purpose of aiding or assisting another member to commit the offence;*
- *(iii) does or omits to do anything for the purpose of aiding or assisting any other abets, counsels, procures or conspires with another member to commit or be a party to an offence; or*
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